

1. Prove $n \log n = \Omega(n)$

Recall the definition:

$$f(n) = \Omega(g(n))$$

if there exist constants $c > 0$ and $n_0 > 0$ such that

$$f(n) \geq c g(n)$$

for all $n \geq n_0$.

Let

$$f(n) = n \log n, g(n) = n.$$

We need:

$$n \log n \geq cn.$$

For $n \geq 2$,

$$\log n \geq 1.$$

Therefore,

$$n \log n \geq n.$$

Choose

$$c = 1, n_0 = 2.$$

Thus,

$$\boxed{n \log n = \Omega(n)}.$$

2. Compare growth rates

$$n, \log n, n \log n, n^2, 2^n, n!$$

arrange them from **slowest to fastest**.

Function	Growth type	Intuition
$\log n$	Logarithmic	Grows extremely slowly
n	Linear	Doubles when n doubles
$n \log n$	Linearithmic	Slightly faster than linear
n^2	Quadratic	n multiplied by itself
2^n	Exponential	Doubles whenever n increases by 1
$n!$	Factorial	Multiplies by an increasing number each step

A useful numerical demonstration (for conceptual understanding, not required for the proof)

For $n = 10$:

$$\begin{aligned}\log_2(10) &\approx 3.3 \\ n &= 10 \\ n \log_2 n &\approx 33 \\ n^2 &= 100 \\ 2^n &= 1,024 \\ n! &= 3,628,800\end{aligned}$$

For $n = 20$:

$$\begin{aligned}\log_2(20) &\approx 4.3 \\ n &= 20 \\ n \log_2 n &\approx 86 \\ n^2 &= 400 \\ 2^n &= 1,048,576 \\ n! &= 2.43 \times 10^{18}\end{aligned}$$

The important asymptotic relationships are:

$$\boxed{\log n = o(n)}$$

$$n = o(n \log n)$$

$$n \log n = o(n^2)$$

$$n^2 = o(2^n)$$

$$2^n = o(n!)$$

So the hierarchy is:

$$\log n < n < n \log n < n^2 < 2^n < n!$$

3. Prove or disprove

Are the following True/False?

A.

$$n^2 = O(n^3)$$

True.

B.

$$n^3 = O(n^2)$$

False.

C.

$$5n + 100 = \Theta(n)$$

True.

D.

$$n \log n = \Theta(n)$$

False.

E.

$$2n^2 + 7n + 1 = \Theta(n^2)$$

True.

F.

$$\log(n^2) = \Theta(\log n)$$

True, because $\log(n^2) = 2 \log n$.